

Analogue filters.

Portal notes:

Active low pass ho

Filter lecture notes (pdf)

Passive ladder ho & hp, bp and bs ho

1. S-plane transfer functions, Q factor and s-plane.
2. Main filter types, Butterworth etc.
3. Passive ladder networks from filter tables
 - a. Low pass
 - b. High pass
4. Active networks from filter tables
 - a. Low pass
 - b. High pass

Questions.

1. What is Q factor?
2. What is the Q factor of a Butterworth Filter?
3. What are poles and zeros?
4. What effect does the position of poles and zeros on the s-plane have on the filter response?
5. For a given order list the four types of filter we covered (butterworth etc) in order of increasingly steep roll off.
6. What is the difference between Cheb 1 and 2 filter implementations?
7. Using the filter tables design a passive filter with a cut-off frequency of 2Mhz and stop band frequency of 4Mhz to give 30dB of attenuation with a 50ohm load and source.
8. The filter in 7 is required to operate for the same performance but at a new cut-off frequency of 6Mhz, re-scale the component values for this new frequency of operation.
9. Repeat question 7 but use an active filter realisation
10. Design a passive high pass filter with a stop band at 2Mhz and Cut-off at 4Mhz to give 20dB attenuation between 50ohm source and load.
11. Repeat question 10 but for an active high pass filter.
12. If designing a band-pass filter how far apart do the two cut-off frequencies need to be before you can simply cascade a low and high pass implementation together?
13. Design a passive and active bandpass filter to meet the following specification:
 - a. Minimum attenuation 20 dB at 50 Hz and 45 kHz
 - b. 3 dB frequencies 100 Hz and 15 kHz
 - c. Gain = 0 dB
 - d. Impedance level to be used for component scaling 10 k Ω .